

Springfield High School

Biology (Grade 9) — Homework Assignment

Name: _____

Roll No: _____

Class: _____ Section: _____

Date: _____

Time: 40 min

Max Marks: 40

Instructions:

- All questions are compulsory.
- Label diagrams clearly.
- Write balanced equations where required.

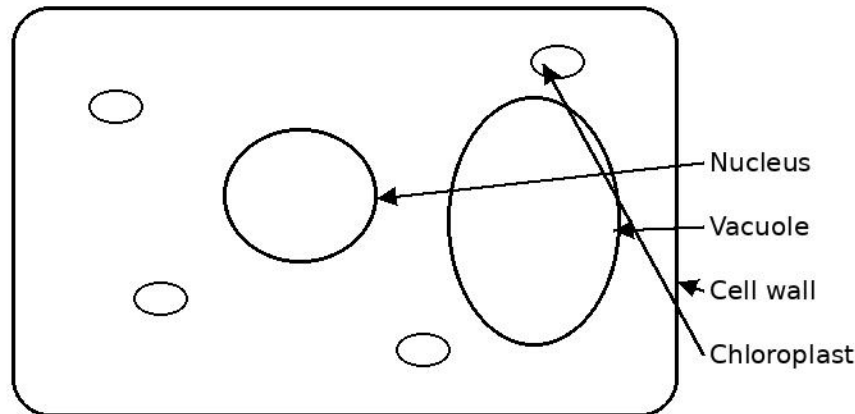
Q1. Write the balanced equation for photosynthesis.

[3]

Ans. $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$

Q2. Draw and label a diagram of a plant cell showing nucleus, vacuole, cell wall and chloroplast.

[5]



Q3. State two functions of the mitochondria.

[3]

Ans. 1) Site of aerobic respiration. 2) Produces ATP (energy) for the cell.

Q4. Complete the observation table for the leaf starch (iodine) test:

[5]

Sample	Light	Iodine colour	Inference
A	Yes	Blue-black	Starch present
B	No	Brown	No starch
C	Partial	Patchy blue	Partial starch

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- Q5.** Differentiate between xylem and phloem (one point each). **[4]**
Ans. Xylem carries water and minerals upward; phloem carries prepared food in both directions.
- Q6.** What is the role of enzymes in digestion? **[3]**
Ans. Enzymes act as biological catalysts that speed up the breakdown of food.
- Q7.** Define osmosis. **[3]**
Ans. Movement of water from higher to lower water potential through a semipermeable membrane.
- Q8.** Write the equation for aerobic respiration. **[3]**
Ans. $C_6H_{12}O_6 + O_2 \rightarrow H_2O + \text{glucose} + \text{energy}$
- Q9.** Name the pores on a leaf surface and state their function. **[3]**
Ans. Stomata — they allow gas exchange and control transpiration.
- Q10.** List two adaptations of a leaf for photosynthesis. **[3]**
Ans. 1) Broad flat surface to absorb more light. 2) Thin structure for easy gas diffusion.
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— End of Biology Paper —